

Event Frequency Model (MKM-EF-001)

Wolfram Oracle Benchmark of Annualisation, Tenor Probability, Compound Loss and Count Families

MKM Research Labs

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1 Executive Summary

This document benchmarks the Event Frequency Model (MKM-EF-001), which turns a gauge's event arrival rate λ and its per-event flood probability p into the annual flood probability every PRS is priced from, and into the multi-year term structure the gauge PRS pricer reads. It asks whether the code computes what its model document (`event_frequency_plan.pdf`) says, and whether what the document says holds. Version 0.1 tested the code against v2.3 of the document and found the term structure understating multi-year risk and annualisation losing precision at small λp. Both are fixed, and the document is issued as v2.4, which this version tests.

The code computes its documented formulae. Over 161 cases — arrival rates from 0.5 to 50 a year, per-event probabilities from 10^{-6} to 0.5, tenors of 1 to 10 years with a constant and a growing rate, three loss catalogues and seven count series — every quantity agrees with a 50-digit implementation of the document's equations to better than 10^{-12} (Table 1). Wolfram and the 50-digit reference agree to 4.9×10^{-30}, so the reference is confirmed by a second, independent implementation. Wolfram's own maximisation of the Negative Binomial likelihood finds nothing higher than the model's fit.

The term structure is exact. Each year now contributes its exceedance intensity $\lambda_y p$, so the probability of a flood within a tenor is $1-\exp(-p\sum_y \lambda_y)$ at every point, to 2.6×10^{-16} (Table 3). v2.3's code compounded each year's annual *probability* as though it were an intensity, which understated that probability by 0.10% to 17.3%, worst at one year and high λp. The correction raises every multi-year probability, and every PRS spread priced from the term structure.

Annualisation keeps its precision. Evaluated as $-\mathrm{expm1}(-\lambda p)$, the annual probability is exact to 1.2×10^{-16} below $\lambda p = 0.0001$, where $1 - \exp(-x)$ lost 10^{-10}.

The document's claims hold (Table 2): the closed form of §5.9 is an identity (Wolfram proves it symbolically), the gate statistic's mean $|z|$ is $\sqrt{2/\pi} = 0.798$, the sampling errors of §7.2 scale as $1/\sqrt{n}$, and the Negative Binomial search reaches its maximum.

Evidence. 13 checks, outcome: all pass (Table 1).

- Wolfram: 15.0.0 for Mac OS X ARM (64-bit) (May 26, 2026) via live kernel session (wolframclient).
- Code under test, at commit `d8acf2c90152`.
- Numerics: numpy 1.26.4, scipy 1.15.1, mpmath 1.4.1.
- Generated 2026-10-27 by `python -m tools.wolfram_oracle ef001`.

2 Model Purpose and Scope

2.1 What is benchmarked

In `src/models/frequency`: `annual_exceedance_probability`, `annual_exceedance_rate` and `annual_hazard_by_year` (`annualise.py`); `term_exceedance_probability` and the rate processes (`rate_process.py`); `analytic_average_annual_loss` and `loss_standard_error`

(`ylt/_reconcile.py`); and the count families, `fit_poisson`, `fit_negbin` and `dispersion_test` (`families/`). With them, `compute_term_structure` in `src/models/hazard/gev.py`, which the per-year hazards feed. Each file is loaded on its own and called as production calls it.

2.2 What is not

The Monte Carlo year simulation is checked by its own reconciliation gate, whose statistic is tested here; the simulation's draws are not. The calibration of λ from gauge records, which the document's §6 and §8 identify as the model's open problem, is not a calculation an oracle can check. This is a benchmark of the implementation (validation question Q8), not independent validation (Q9).

3 Mathematical Framework

With $N \sim \text{Poisson}(\lambda)$ events a year, each flooding independently with probability p, the document's §5.9 gives

$$\mathbb{E}\big[(1-p)^N\big] = e^{-\lambda p}, \qquad P(\text{at least one flood in a year}) = 1 - e^{-\lambda p}, \tag{1}$$

and the aggregate rate λp, which is never smaller. Over a tenor of T years with rates $\lambda_0, \ldots, \lambda_{T-1}$ (constant, or growing geometrically as $\lambda_0(1+g)^y$), the exact tenor probability is $1 - \exp\big(-p\sum_y \lambda_y\big)$.

The term structure (v2.4, §5.10). The hazard builder hands `compute_term_structure` each year's intensity $\lambda_y p$, which it accumulates, so each point is the exact tenor probability. v2.3's code handed it the year's *probability* $q_y = 1 - e^{-\lambda_y p}$ instead, giving $1 - \exp\big(-\sum_y q_y\big)$; since $q_y < \lambda_y p$ that always understated, by more as λp grows, and most relative to the answer at one year.

Compound loss. With event weights w_k and losses L_k, the annual loss is compound Poisson, with mean $\lambda \sum_k w_k L_k$ and, over n simulated years, standard error $\sqrt{\lambda \sum_k w_k L_k^2 / n}$.

Count families. The dispersion statistic $(n-1)s^2/\bar{x}$ is χ^2_{n-1} under Poisson; the Negative Binomial is fitted with its mean at $\bar{x}$ and its dispersion $\alpha = 1/r$ by bounded search, which is the joint maximum likelihood because the Negative Binomial's estimating equation for the mean is solved by $\bar{x}$ at every r.

4 Benchmark Method

The equations above are evaluated with `mpmath` at 50 significant digits, from the document rather than the code. Wolfram evaluates them independently from each input's exact value at 60-digit precision: the compound loss through `CompoundPoissonDistribution` of an `EmpiricalDistribution` (its rate scaled by the weights' exact total, which `EmpiricalDistribution` would otherwise renormalise away), the p-values through `ChiSquareDistribution`, and the Negative Binomial maximum by its own `FindMaximum` over both parameters, with the Poisson limit as the supremum for under-dispersed data. It proves the identity of §5.9 with `Expectation` and `Simplify`. Every annual probability is judged against the 10^{-12} tolerance, down to $\lambda p = 5 \times 10^{-7}$.

5 Validation and Backtesting

Table 1: Benchmark checks

Check	Result	Tolerance	Outcome
Annual probability $1 - e^{-\lambda p}$, λp from 5×10^{-7}: model vs reference	1.3×10^{-16}	1.0×10^{-12}	pass
Aggregate rate λp: model vs reference	5.6×10^{-17}	1.0×10^{-12}	pass
Tenor probability $1 - e^{-p \sum \lambda_y}$: model vs reference	2.6×10^{-16}	1.0×10^{-12}	pass
Term structure, compounded year by year (§5.10): model vs reference	2.6×10^{-16}	1.0×10^{-12}	pass
Compound-Poisson average annual loss: model vs reference	2.3×10^{-16}	1.0×10^{-12}	pass
Its standard error over n years: model vs reference	4.6×10^{-17}	1.0×10^{-12}	pass
Dispersion statistic $(n-1)s^2/\bar{x}$: model vs reference	1.4×10^{-16}	1.0×10^{-12}	pass
Over-dispersion p-value (χ^2 survival): model vs reference	9.4×10^{-15}	1.0×10^{-10}	pass
Under-dispersion p-value (χ^2 CDF): model vs reference	5.5×10^{-16}	1.0×10^{-10}	pass
Poisson log-likelihood: model vs reference	1.3×10^{-15}	1.0×10^{-12}	pass
Negative Binomial log-likelihood at its maximum: model vs reference	1.3×10^{-12}	1.0×10^{-8}	pass
Wolfram agrees with the reference (closed forms, χ^2, Poisson)	4.9×10^{-30}	1.0×10^{-12}	pass
Wolfram's own Negative Binomial maximum finds no higher likelihood	0	1.0×10^{-8}	pass

5.1 The model document's claims

Table 2: The model document's claims (v2.4), tested

Claim	Stated	Benchmark finds	Outcome
§5.9 closed form	$E[(1-p)^N] = e^{-\lambda p}$; aggregate $\geq$ occurrence	identity confirmed symbolically; aggregate $\geq$ occurrence in every case	met
§5.9 gate statistic	mean $\lvert z \rvert = \sqrt{2/\pi} = 0.798$	0.797885	met
§7.2 sampling error	5, 1.6, 0.5, 0.16% at $10^3..10^6$ years	$1/\sqrt{n}$ from the first gives 5.00%, 1.58%, 0.50%, 0.16%	met
§5.10 term structure	$1 - \exp(-p \sum_y \lambda_y)$ at every tenor	to 2.6×10^{-16}; v2.3 understated it by 0.10–17.3%	met
C21 precision	$-\text{expm1}(-\lambda p)$ keeps full precision	1.2×10^{-16} relative below $\lambda p = 0.0001$ (v2.3: 1.0×10^{-10})	met
Negative Binomial search	bounded scalar search on α	log-likelihood within 1.8×10^{-10} of the maximum	met

The §7.2 errors imply a relative error $\sqrt{(1-q)/(nq)}$ with $q \approx 0.29$: the table quotes one subject's figures, not a bound.

5.2 The multi-year term structure

Table 3: The multi-year exceedance probability: the term structure now, and v2.3's convention

Rate	λ_0	p	T	Exact	Now	v2.3	v2.3 gap
const	2	0.001	1	0.001998	0.001998	0.001996	-0.10%
const	2	0.001	10	0.019801	0.019801	0.019782	-0.10%
const	2	0.01	1	0.019801	0.019801	0.019607	-0.98%
const	2	0.01	10	0.181269	0.181269	0.179641	-0.90%
const	2	0.05	1	0.095163	0.095163	0.090775	-4.61%
const	2	0.05	10	0.632121	0.632121	0.613887	-2.88%
const	10	0.001	1	0.009950	0.009950	0.009901	-0.50%
const	10	0.001	10	0.095163	0.095163	0.094712	-0.47%
const	10	0.01	1	0.095163	0.095163	0.090775	-4.61%
const	10	0.01	10	0.632121	0.632121	0.613887	-2.88%
const	10	0.05	1	0.393469	0.393469	0.325288	-17.33%
const	10	0.05	10	0.993262	0.993262	0.980448	-1.29%
trend	5	0.001	1	0.004988	0.004988	0.004975	-0.25%
trend	5	0.001	10	0.060953	0.060953	0.060764	-0.31%
trend	5	0.01	1	0.048771	0.048771	0.047600	-2.40%
trend	5	0.01	10	0.466819	0.466819	0.456195	-2.28%
trend	5	0.05	1	0.221199	0.221199	0.198443	-10.29%
trend	5	0.05	10	0.956910	0.956910	0.932189	-2.58%

6 Model Limitations and Known Weaknesses

1. **Benchmark, not validation.** The code is shown to compute the documented formulae; whether λ is right for a gauge is the question the document's own §8 raises and leaves open.

2. **Stored term structures.** Term structures already written by the hazard builder, and any spreads priced from them, carry v2.3's understatement until the hazard curves are rebuilt.

3. **Naming.** The hazard curve still stores the annual *probability* under `annual_hazard_rate_*`; PRS pricing reads it as a probability, so the value is right and only the name misleads.

7 Document History

Date	Version	Description	Author
26-Oct-2026	0.1	First issue: tests v2.3; finds the term structure understated	David K Kelly
27-Oct-2026	0.2	After the fixes: tests v2.4	David K Kelly